

Joseph H. Rudoler

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Education

University of Pennsylvania

- 2023 – **Ph.D., Statistics and Data Science**, *The Wharton School*
Advisor: Giles Hooker
- 2020 – 2023 **M.S.E., Data Science**, *School of Engineering and Applied Science*
Thesis title: Transfer learning to improve memory state classification from EEG. Thesis advisors: Konrad Kording and Michael Kahana
- 2016 – 2020 **B.A., Physics and Astronomy (*Magna Cum Laude*)**, *School of Arts and Sciences*
Minors: English, Music

Employment

Computational Memory Lab (PI: Michael Kahana), U. of Penn.

- 2021 – 2023 **Data and Programming Specialist**
Staff data scientist, managing data pipelines, software development
- 2020 – 2021 **Clinical Research Specialist**
Coordinating multi-site project, collecting iEEG data from epilepsy patients, maintaining clinical imaging and electrode localization pipeline

Radian Group, Inc.

- 2019 **Pricing Intern**
Developing pricing models for mortgage insurance premiums, efficient data processing, dashboards / data visualizations

Teaching

- 2026 **STAT4730/7730, Data Science with ChatGPT**
Teaching Assistant
Undergraduate and MBA course on data analysis with LLMs.
- 2025 – **STAT4010, Sports Analytics Capstone**
Teaching Assistant
Undergraduate capstone in statistical analysis of sports data.
- 2025 **US-Israel Academic Bridge Fellowship: Data Science Track**
Instructor
Designed and taught a 5-week summer course in statistics and data science for Israeli undergraduate exchange students. New curriculum emphasizes sampling variability and computational approaches.
Course site: <https://jrudoler-teaching.github.io/understanding-uncertainty/>
- 2023 – **STAT6130, Regression Analysis for Business**
Teaching Assistant
MBA core curriculum course on statistical foundations and regression
- 2023 – 2025 **Wharton Moneyball Academy**
Lecturer
Summer course for high school students in sports analytics and data science

Teaching (continued)

- 2021 **Estes Summer Workshop in Model-based Cognitive Electrophysiology**
Lecturer
2-week summer course, mostly for doctoral and postdoctoral researchers, in analyzing EEG data to study behavior. Wrote new lecture materials focusing on analyzing periodic and aperiodic components of EEG.
- 2019 – 2020 **CIS105, Computational Data Exploration**
Tutor (individual)
- 2017 – 2019 **PHYS150/151, Physics I (Mechanics) and II (Electricity & Magnetism)**
Tutor (open office hours)

Research

In Preparation

- 1 **Rudoler, J. H.**, Asemota, A., & Hooker, G. (2026). *Targeted learning for inference on data fairness* [Manuscript in preparation].
- 2 **Rudoler, J. H.**, DeHaan, R., Kording, K. P., & Dobriban, E. (2026). *Do we decode better than chance? Efficient hypothesis testing for neural decoders* [Manuscript in preparation].

Preprints

- 1 **Rudoler, J. H.**, Tan, K., Tessler, B., Kong, T., & Boix Adserà, E. (2026, September 28). *Improving scalable oversight with co-trained monitors*. arXiv: 2609.36049 [cs.LG].
<https://doi.org/10.48550/arXiv.2609.36049>

Peer-Reviewed Papers

- 1 **Rudoler, J. H.**, Saxe, A. M., & Kording, K. (2026). Weight space gradient descent induces approximate activity descent, making learning rules testable. *NeurIPS 2026 Workshop on Symmetry and Geometry in Neural Representations (NeurReps), Extended Abstract Track*. [Accepted].
- 2 **Rudoler, J. H.**, Tan, K., Hooker, G., & Kording, K. P. (2026). Estimating Implicit Regularization in Deep Learning. *NeurIPS 2026*.
<https://doi.org/10.48550/arXiv.2605.05436> [Accepted].
- 3 Dougherty, M. R., Chang, W., **Rudoler, J. H.**, Kateman, B. S., Halpern, D. J., Bruska, J. P., Diamond, N. B., & Kahana, M. J. (2024). Neural correlates of memory in an immersive spatiotemporal context. *Journal of Experimental Psychology: Learning, Memory, and Cognition*.
<https://doi.org/10.1037/xlm0001341>
- 4 Kahana, M. J., Lohas, L. J., Healey, M. K., Aka, A., Broitman, A. W., Crutchley, P., Crutchley, E., Alm, K. H., Kateman, B. S., Miller, N. E., Kuhn, J. R., Li, Y., Long, N. M., Miller, J., Paron, M. D., Pazdera, J. K., Pedisich, I., **Rudoler, J. H.**, & Weidemann, C. T. (2024). The Penn Electrophysiology of Encoding and Retrieval Study. *Journal of Experimental Psychology: Learning, Memory, and Cognition*. <https://doi.org/10.1037/xlm0001319>
- 5 **Rudoler, J. H.**, Bruska, J. P., Chang, W., Dougherty, M. R., Kateman, B. S., Halpern, D. J., Diamond, N. B., & Kahana, M. J. (2024). Decoding EEG for optimizing naturalistic memory. *Journal of Neuroscience Methods*, 410, 110220.
<https://doi.org/10.1016/j.jneumeth.2024.110220>
- 6 **Rudoler, J. H.**, Herweg, N. A., & Kahana, M. J. (2023). Hippocampal Theta and Episodic Memory. *Journal of Neuroscience*, 43(4), 613–620.
<https://doi.org/10.1523/JNEUROSCI.1045-22.2022>

Presentations

- 1 **Rudoler, J. H.** *Do We Decode Better than Chance? Efficient Hypothesis Testing for Neural Decoders*. Cognitive Computational Neuroscience, Amsterdam, Netherlands, 2025. (Poster).
- 2 **Rudoler, J. H.** *Decoding and Optimizing Episodic Memory*. Cognitive Neuroscience Society, San Francisco, California, USA, 2022. (Poster).
- 3 **Rudoler, J. H.** *Decoding and Optimizing Episodic Memory*. MathPsych, Toronto, Canada, 2022. (Poster).
- 4 **Rudoler, J. H.** *Oscillatory and Fractal Biomarkers of Human Memory*. Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal, 2022. (Poster).

Datasets

- 1 Kahana, M. J., **Rudoler, J. H.**, Lohnas, L. J., Healey, K., Aka, A., Broitman, A., Crutchley, E., Crutchley, P., Alm, K. H., Katerman, B. S., Miller, N. E., Kuhn, J. R., Li, Y., Long, N. M., Miller, J., Paron, M. D., Pazdera, J. K., Pedisich, I., & Weidemann, C. T. (2023). *Penn Electrophysiology of Encoding and Retrieval Study (PEERS)*. OpenNeuro. <https://doi.org/10.18112/openneuro.ds004395.v2.0.0>
- 2 **Rudoler, J. H.**, Dougherty, M. R., Katerman, B. S., Bruska, J. P., Chang, W., Halpern, D. J., Diamond, N. B., & Kahana, M. J. (2023). *"Spatial memory and non-invasive closed-loop stimulus timing"*. OpenNeuro. <https://doi.org/10.18112/openneuro.ds004706.v1.0.0>

Awards, Funding, and Certifications

Awards and Honors

- 2026 **Donald S. Murray Prize for Excellence in Teaching**
Statistics and Data Science Department, The Wharton School.
- 2023 **Honorable Mention for Best Thesis**, M.S.E. in Data Science.
- 2022 **Finalist, NFL Big Data Bowl**, team competition in sports analytics.
- 2017 – 2020 **Dean's List**, U. of Penn.

Funding and Fellowships

- 2026 **Wharton AI & Analytics Initiative Funding**
Teaching statistics and data science from first principles: a data-oriented, AI-enabled course (with Adi Wyner).
- 2025 – 2026 **Tel Aviv University Research Fellowship for Outstanding International Students**
Visiting research in the Blavatnik School of Computer Science and AI.
- 2019 **Jeffrey Greenberg Undergraduate Research Fellowship**, U. of Penn.

Certifications

- 2025 **Certificate in College and University Teaching**
Awarded by Penn's Center for Excellence in Teaching, Learning, and Innovation.